

How Humans Should Converse with AI

A Problem-First Adaptive Dialogue Conversion Protocol for Durable Human Capability

From live problem selection to epistemic expansion, Human Return, action, and world feedback

Yaoharee Lahtee

Human-AI Readout Programme; ARAYA Nikah Social Enterprise Co., Ltd., Bangkok, Thailand

ORCID 0009-0005-3861-0626 • v1.3 — glosa-checked edition of v1.2, 6 September 2026 original

Status. K0 theory, problem-first adaptive protocol, and empirical research agenda. This v1.2 revision uses v1.1 as its anchor and adds the missing topic-selection layer before triage. The canonical eight-stage Dialogue Conversion Protocol is preserved, but human-development use now begins from a live problem: a presently consequential unresolved difference in the person's life, work, understanding, or environment. Problem-first is a default, not a monopoly; curiosity, play, art, basic inquiry, and routine delegation remain legitimate entry modes. The revision also closes the loop after Human Return through action and world feedback, so AI dialogue is judged not only by what becomes thinkable or retained but, where relevant, by whether revised understanding survives contact with the situation that generated the problem. Licence CC BY 4.0.

Edition note. This is glosa-checked edition v1.3 of the founder's v1.2 (6 September 2026 original). Changes are limited to: (1) citation-lineage repair — every internal Human-AI Readout Programme self-citation now carries its deposited Zenodo DOI and the actually-deposited version string; Table 1's six footnotes that pointed to a single miscellaneous internal reference are split into their own correctly-cited entries, and its two rows with no separately deposited source are marked as programme working notes; (2) renaming §2.1 to "Contribution boundary" and rewording four priority-adjacent phrases in it. No section, equation, table, proposition, or reference was added, removed, renumbered in content, or otherwise reworded. The v1.2 PDF remains the anchor document of record; this edition is a citation-accurate copy for deposit, not a superseding claim.

Central claim. A human-AI conversation should be evaluated by the human state it produces after the conversation, but the conversation should also begin from an agenda that remains meaningfully human-owned. For learning, judgment, problem solving, or human development, the default topic should therefore originate in a live human problem rather than an AI-supplied agenda. The outer architecture is Topic Entry → Triage → Minimum Sufficient Dialogue → Human Return → Action/World Feedback; inside Standard and Critical use, the canonical eight stages remain Anchor → Expand → Oppose/Boundary-Test → Discriminate → Verify → Integrate → Remove → Return. AI is used aggressively for search, divergence, translation, recombination, counterexample, and critique; humans retain agenda/problem ownership, normative responsibility, world contact, integration, and final accountable judgment. The practical design objective remains minimum sufficient friction: enough resistance to preserve learning, verification, fallback, and agency, but not so much burden that users abandon the protocol or automate trivial work inefficiently.

ABSTRACT

Artificial intelligence increasingly mediates not only answers but attention: it can help determine what users ask about next, which interpretations become salient, and which problems are treated as worth solving. A protocol that begins only after a prompt therefore starts too late. This paper elevates the adaptive Dialogue Conversion Protocol (DCP) by adding a Topic Entry Condition before triage. For learning, judgment, problem solving, and human development, the default topic should arise from a live problem: a presently consequential unresolved difference in the person's life, work, understanding, or environment. The rationale is architectural rather than therapeutic: problem-first dialogue preserves human agenda ownership, bounds relevance, provides routes to world-side feedback, and makes durable Human Return more observable. Yet problem-first is not problem-only; curiosity, play, art, basic inquiry, and routine delegation remain protected entry modes.

The v1.1 adaptive deployment layer is retained: DCP-Lite uses State, Challenge, Check, Own for ordinary low-

stakes use; DCP-Standard uses the full Anchor, Expand, Oppose/Boundary-Test, Discriminate, Verify, Integrate, Remove, Return architecture when learning, research, judgment transfer, or future competence matters; DCP-Critical adds independent verification, escalation, and accountable professional/world-side judgment. The revision further closes the human-development loop after Return through Action and World Feedback, yielding a problem-conditioned cycle: Live Problem → Question → Dialogue → Human Return → Action → World Feedback → revised understanding or a new problem.

The resulting proposal is a Problem-First, Adaptive Dialogue Conversion Protocol. Its governing principles are Human Agenda Ownership, Minimum Sufficient Friction, World Answerability, and Open Exploration Preserved. The paper proposes new empirical tests comparing live-problem-first, AI-suggested-topic, and curiosity-led dialogue while holding task difficulty and assistance constant. The central prediction is that, when durable human development is the goal, problem-first topic selection will often improve relevance, ac-

tion transfer, and Human Return without requiring that all worthwhile inquiry begin from practical problems.

Keywords: human-AI dialogue; problem-first inquiry; problem framing; topic selection; epistemic agency; Human Return; AI-assisted learning; overreliance; critical thinking; curiosity; metacognition.

1 INTERNAL PROGRAMME LITERATURE REVIEW: WHERE THE PROTOCOL COMES FROM

The protocol is not introduced as a free-standing list of prompt tips. It is the operational convergence of a sequence of prior papers in the Human-AI Readout Programme. The sequence matters because each paper solved a different failure mode that becomes visible once dialogue is treated as a history-shaped epistemic process rather than as independent prompt-response pairs.

The literature converges on a role-separated architecture. The human enters with a retained state—experience, assumptions, stakes, partial knowledge, and unresolved differences—but exports only a bounded articulation. AI receives the articulation, not the whole person. Repeated turns can then alter what becomes easier to think next. The key question is not whether AI produces more content, but whether dialogue enlarges a person’s reachable, discriminable, corrigible space of inquiry and whether that enlargement survives the session boundary [1, 2].

2 EXTERNAL LITERATURE: CONVERGENCE AND THE REMAINING GAP

External research independently supports several pieces of this architecture while also showing why a full dialogue protocol is still missing.

First, problem construction and problem framing are themselves consequential cognitive operations, not neutral preliminaries. Reiter-Palmon reviews evidence that how a problem is constructed influences creative production, while recent innovation research treats problem framing as a dynamic creative process in its own right rather than a fixed input to solution generation [13, 14]. This supports a narrower claim than “all knowledge begins from problems”: when an agent seeks to learn, decide, repair, or act, selecting and framing the live problem is part of the epistemic work. The topic of AI dialogue is therefore not a free variable.

Second, human-AI combination is not automatically synergistic. Vaccaro, Almaatouq, and Malone meta-analyzed 106 experiments and found average augmentation relative to humans alone but no average synergy relative to the better of human or AI; decision tasks performed particularly poorly compared with creation tasks [15]. This makes “human+AI” an insufficient success criterion.

Third, overreliance is a design problem, not merely an indi-

vidual defect. Buçinca, Malaya, and Gajos show that cognitive forcing functions can reduce overreliance, although users may prefer less effortful designs [16]. Vasconcelos et al. further show that reliance depends on the cost-benefit structure of engaging with AI explanations, while de Jong et al. find that partial explanations can reduce overreliance under some conditions [17, 18]. Recent CHI work on AI-assisted critical thinking similarly reports reduced overreliance but increased cognitive load, implying a real friction-performance trade-off rather than a free benefit [19].

Fourth, learning science predicts that easy access to complete solutions can conflict with durable learning. The generation effect, retrieval practice, and productive-failure literatures show that attempting, retrieving, reconstructing, and even failing productively can improve later retention or transfer compared with passive presentation or immediate instruction [20–24]. This is not evidence that AI must always withhold answers; it is evidence that the timing and form of assistance can alter what the human retains.

Fifth, contemporary AI evidence shows that this distinction is empirically consequential. Bastani et al. report that generative-AI access can sharply improve practice performance while unguarded use reduces later unaided performance after the AI is removed; guardrailed tutoring mitigates the loss [25]. Glickman and Sharot show that repeated interaction with biased AI can alter later human judgments, often without users fully recognizing the influence [26]. Buçinca et al. argue explicitly for optimizing human-centric objectives such as learning rather than accuracy alone [27].

The residual gap is therefore narrow and important: the literature contains mechanisms of problem framing, overreliance, learning, explanation, task-specific synergy, and feedback, but it lacks a general conversation-level architecture that asks which topic deserves attention before the prompt, distinguishes AI-suggested agenda from human-owned live problems, separates expansion from verification, introduces deliberate opposition and equivalence testing, and ends with AI removal plus an unaided Human Return audit and, where action is relevant, renewed world feedback.

2.1 Contribution boundary

No individual component of the protocol is claimed as a new contribution; each has substantial predecessors, cited above. Generation, retrieval practice, productive failure, cognitive forcing, explanation, adversarial critique, source verification, and reflection each have substantial predecessors. The residual contribution is their ordered integration around a different terminal outcome: the human state after decisive AI support is removed. Six features are proposed as the distinctive architectural combination: (i) a topic-entry distinction between live problems, open exploration, and routine delegation; (ii) a pre-prompt human baseline; (iii) explicit discrimination of non-equivalent alternatives rather than raw answer count; (iv) a mandatory transition from AI-mediated expansion to partially independent world-side verification; (v) an AI-removal bound-

Table 1: Internal programme synthesis. The present protocol retains the result in the middle column and operationalizes it in the right column.

Prior work	Retained result	Role in the Dialogue Conversion Protocol
When AI Expands Human Potential	AI can expose alternatives, hidden assumptions, and productive dissonance, but expansion requires corrigibility, route-openness, criticism, and world-answerability; fluent output is not human growth.	Supplies the target: disciplined revision rather than convenience or throughput. [3]
The Readout Condition	Epistemically weight-bearing distinctions require existence, attribution, provenance, and disclosure of augmentation; a system cannot self-certify truth merely by producing a coherent claim.	Supplies provenance typing and the no-self-certification rule. [4]
From Problem to Hypothesis	Practical inquiry begins from a retained problem: a consequential mismatch requiring explanation, repair, or decision; imagination then enlarges a bounded reachable hypothesis space.	Supplies the Topic Entry Condition, Problem-First Dialogue Principle, non-equivalent alternative generation, and semantic-basin checking. [5]
The Epistemic Chain Reaction Candidate Forgetting	The human retains the problem/question while AI decomposes, transports, branches, critiques, and produces readout-distinguishable candidate routes. Knowledge-like form can be mistaken for validated knowledge; provisional candidates harden when status is forgotten.	Supplies controlled expansion and recursive branching. [6] Supplies candidate labels and mandatory re-verification before load-bearing use (programme working notes, not separately deposited).
Experience Is the Human LoRA	Exposure is not retention. Some residues alter later human interpretation/action; others pass through.	Supplies the session-boundary retention question. [7]
AI, Translation, and Event-Specific Context	AI reorganizes recorded interpretations and context access but does not directly reconstruct event reality.	Supplies the distinction between language mediation and target-world access. [8]
Human Learning as Epistemic Architecture	Concepts and tools become usable through problem-context, deliberate practice, real problems, feedback, and revision.	Supplies problem-context binding, Tool/Skill Return, and the Action → World Feedback closure. [9]
Context Wisdom / Translation of Reality	Meaning is context-sensitive and interpretation-bounded; repeated relations can become easier to traverse without becoming more warranted.	Supplies context-checking and semantic-momentum caution (programme working notes, not separately deposited).
Epistemic Fusion or Epistemic Tunnel	Repeated Human-AI dialogue can expand or narrow future accessible routes; amplification is direction-neutral; resistance and selective retention determine epistemic direction.	Supplies precommit resistance, stop/reset rules, and the expansion-versus-tunnel diagnostic. [1]
Choice Begins Before Choice	Humans do not choose from all formal possibilities but from a history-shaped live field; AI can reshape which alternatives become thinkable before the next prompt.	Supplies the pre-prompt human state and live-possibility audit. [2]
CTSA Human Return Readout After Labour / Human Conversion Imperative	Coupled performance is not human capability. Durable return must be demonstrated unaided as Concept, Tool, Skill, or Alternative Return. Human potential also depends on exit, material claim, social reproduction, and reversibility; maximum AI use is not maximum human conversion.	Supplies the terminal Remove/Return test. [10] Supplies system-level constraints and urgency where dependency becomes hard to reverse. [11, 12]

ary followed by a typed Human Return audit; and (vi) closure, where relevant, through action and returned world feedback. If this ordering adds no predictive or intervention value beyond simpler critical-thinking instructions, the protocol should be compressed rather than defended by terminology.

3 THE UNIT OF ANALYSIS: A CONVERSATION CHANGES THE NEXT CONVERSATION

A prompt is not the starting state. Let $H_{s,0}$ denote the human state entering session s , containing retained experience, current meanings, values, unresolved differences, recent interaction

history, and a live-possibility profile. Linguistic articulation exports only a bounded question:

$$H_{s,0} \xrightarrow{L_H} Q_{s,0}. \quad (1)$$

AI responds to Q , not to the entire human state. The response becomes part of the human's next interpretive environment:

$$Q_{s,t} \rightarrow AI_{s,t} \rightarrow Y_{s,t} \xrightarrow{R_H^{AI}} E_{H,s,t}. \quad (2)$$

If some residue is retained, then

$$H_{s+1,0} = U_H(H_{s,0}, E_{H,s,*}^{AI}, \delta^{\text{world}}, X^{\text{other}}), \quad (3)$$

so the next session can begin from a changed reader. This is why conversational quality cannot be judged only by the current answer. Repeated interaction may make objections easier to formulate or harder to imagine, widen or narrow live possibilities, and improve or erode unaided skill [1, 2, 26].

4 TOPIC ENTRY CONDITION: CHOOSE THE CONVERSATION BEFORE OPTIMIZING IT

A dialogue protocol that begins with Anchor still assumes that some topic has already captured the user’s attention. In an AI-rich environment this assumption becomes unsafe: systems can suggest topics, recommend goals, surface concerns, and recursively shape what feels worth asking next. Topic selection is therefore an agency problem before it is a prompting problem.

For human-development use, define a live problem as

$$P_{H,t}^{\text{live}} = \text{a presently consequential unresolved difference} \quad (4)$$

in the person’s life, work, understanding, relationship to evidence, or environment. “Problem” is used broadly: it can be a failure, mismatch, decision, unexplained recurrence, missing capability, conflict between expectation and observation, or a question whose unresolved state matters enough to justify attention. This is consistent with the programme’s earlier move from generic observation to a retained problem that demands explanation, repair, or decision [5].

The Topic Entry Condition separates three legitimate entry modes:

$$\text{TopicEntry} \in \{\text{LiveProblem}, \text{OpenExploration}, \text{RoutineDelegation}\}. \quad (5)$$

LiveProblem is the default when the goal is learning, judgment, problem solving, behavior change, or human development. OpenExploration protects curiosity, play, art, basic science, speculative inquiry, and interests whose value is not yet tied to a practical mismatch. Curiosity research itself treats exploration as an adaptive process driven by uncertainty, information, and learning progress rather than as wasted attention [28]. RoutineDelegation covers low-value or repetitive tasks where the user need not convert every operation into durable human skill.

Problem-First Dialogue Principle [Open]. When AI is used for human development rather than entertainment or routine execution, the default topic should originate in a live human problem rather than an AI-supplied agenda. The reason is fourfold:

$$\text{Problem-First} \Rightarrow \begin{cases} \text{Human Agenda Ownership,} \\ \text{Bounded Relevance,} \\ \text{World-Side Testability,} \\ \text{Observable Human Return.} \end{cases} \quad (6)$$

The claim is not that every problem is correctly framed. Problem construction can itself be creative, contested, and revisable

[13, 14]. AI may help elicit or reframe a problem, but any reframing that changes the agenda should be made visible to the user rather than silently replacing the original concern.

Problem-first is not problem-only. A universal requirement that every conversation justify itself by present utility would destroy exploration. The protocol therefore preserves an explicit exploration gate:

$$\text{Problem-First} \neq \text{Problem-Only}. \quad (7)$$

The relevant governance question is not “Is this topic useful now?” but “Who selected the topic, why is attention being spent here, and does the user still recognize the difference between a self-owned inquiry and an AI-supplied agenda?”

A compact Topic Entry prompt is: “What unresolved difference in my present life, work, understanding, or environment deserves attention now? If I am exploring from curiosity rather than solving a problem, keep that mode explicit rather than inventing a problem for me.”

5 DEPLOYMENT LAYER: TRIAGE AFTER TOPIC ENTRY

The canonical eight-stage DCP is a protection architecture, not a requirement that every user perform a mini peer review before asking for a translation or formatting change. Real-life deployment begins by asking what must be preserved.

Let the deployment policy depend on

$$\pi^{\text{deploy}} = f(\text{Stakes}, \text{LearningNeed}, \text{Irreversibility}, \text{DependencyRisk}, \text{UserSkill}). \quad (8)$$

The proposed triage has three modes.

DCP-Lite: ordinary low-stakes use.

$$\text{State} \rightarrow \text{Challenge} \rightarrow \text{Check} \rightarrow \text{Own} \quad (9)$$

State: what am I trying to do? Challenge: what obvious thing might I be missing? Check: which fact, if wrong, would materially change the action? Own: do I understand and accept the final output well enough to use it responsibly? Lite is appropriate for routine writing, low-cost planning, formatting, ordinary translation, and reversible decisions where durable learning is not the main goal.

DCP-Standard: capability-sensitive use. Use the full eight-stage architecture when learning, research, analytical judgment, transfer to later unaided work, or future fallback competence matters.

DCP-Critical: consequential or hard-to-reverse use. Use the full architecture plus at least one independent evidence/human route and an escalation rule. AI may organize evidence, generate alternatives, or stress-test reasoning, but it is not the sole certifier of health, legal, financial, safety-critical, or other high-stakes conclusions. This yields the deployment rule

$$\begin{aligned} \text{Topic Entry} &\rightarrow \text{Triage} \rightarrow \text{Minimum Sufficient Dialogue} \\ &\rightarrow \text{Escalate if Stakes Rise} \rightarrow \text{Periodic Return.} \end{aligned} \quad (10)$$

The governing principles are now paired. Human Agenda Ownership governs what enters the dialogue; Minimum Sufficient Friction governs how demanding the dialogue should become. Protective effort should rise with learning value, failure cost, irreversibility, and dependency risk, but fall when the task is trivial, reversible, and not capability-sensitive.

Adaptive-friction proposition [Open]. The best real-world protocol will often be the least burdensome policy that preserves a predeclared minimum of task safety, human understanding, verification, and fallback competence; maximum friction is not the objective.

6 THE DIALOGUE CONVERSION PROTOCOL

The proposed protocol is

Anchor → Expand → Oppose → Discriminate
→ Verify → Integrate → Remove → Return. (11)

The sequence is not a ritual. Each stage protects against a distinct collapse.

6.1 Anchor: stabilize the human-owned topic before AI expansion

Once the Topic Entry Condition identifies a live problem or explicit exploration goal, the human records a minimal pre-AI anchor:

$$A_0 = \langle P_0, M_0, U_0, E_0, F_0, S_0 \rangle, \quad (12)$$

where P_0 is the problem as currently understood, M_0 the current model or tentative answer, U_0 the important unknowns, E_0 available evidence, F_0 a condition that would change the human's mind, and S_0 the stakes/domain.

The anchor need not be sophisticated. Its purpose is to create a human baseline against which model change and Human Return can later be measured. In learning tasks, a first unaided attempt may itself be productive because generation and productive-failure research show that initial effort can improve later conceptual learning and transfer [20, 23, 24].

Anchor elicitation for novices. Some users come to AI precisely because they do not yet possess a coherent M_0 . The protocol must not convert epistemic literacy into an entry requirement. In that case AI first asks a small number of clarification questions rather than demanding a model: “What are you trying to understand or decide?”, “What do you know with reasonable confidence?”, and “What is the most important uncertainty?” The human-approved answer to those questions becomes the anchor. Thus

$$AI_{\text{clarify}} \rightarrow \{\text{Known, Unknown, Goal, Stakes}\} \rightarrow \text{Human Anchor.} \quad (13)$$

AI may help elicit the anchor, but it should not silently own the topic or problem definition. If clarification changes the original agenda, that change should be surfaced for human acceptance.

Prompt function: “Here is my current model and what I do not know. If I do not yet have a model, help me clarify the problem first. Do not merely improve my view; identify what would most strongly challenge it.”

6.2 Expand: use AI as a divergence engine

AI is then used for the tasks in which it has comparative leverage: search, translation, decomposition, analogy, recombination, counterexample, cross-domain transport, and candidate generation. The objective is not raw idea count but an increase in readout-distinguishable candidate classes:

$$Q_t \xrightarrow{T_{AI}} \{Q_{i,t+1}, K_{i,t}\}_{i=1}^n. \quad (14)$$

AI outputs remain knowledge-like candidates unless independently warranted. A useful expansion request asks for several structurally different routes: opposite causal direction, missing-variable model, cross-domain analogy, boundary-case explanation, and the strongest alternative from a rival framework.

Prompt function: “Generate three non-equivalent explanations. For each, state which assumption changes, what evidence would favor it, and what would make it fail.”

6.3 Oppose or boundary-test: precommit resistance without false balance

Before the dialogue converges, the human should expose the leading model to resistance. This stage is intentionally placed early because once a semantic basin becomes fluent, later objections may be absorbed as local patches rather than treated as genuine tests. Cognitive-forcing research supports the broader design principle that forcing engagement can reduce overreliance, even when users prefer easier interfaces [16, 18].

However, opposition is evidence-sensitive. The protocol must not manufacture a symmetrical “other side” when the evidence is strongly asymmetric. It therefore distinguishes two operations:

Rival Generation : for genuinely open spaces, (15)

Boundary Test : for strongly asymmetric evidence. (16)

Rival Generation asks for the strongest non-equivalent hypothesis, counterexample, falsifier, missing comparison, or failure mode. Boundary Testing asks where the established claim stops applying, which uncertainty remains, what confounder or measurement limit matters, and what observation would legitimately revise confidence. Hence

Oppose a claim $\neq$ manufacture an opposite claim. (17)

AI critique is not independent validation, but it can expose where independent validation is needed.

Prompt function: “If this is genuinely uncertain, build the strongest rival and the decisive discriminating test. If the evidence is already strongly one-sided, do not create false balance; identify boundary conditions, uncertainty, confounders, and legitimate revision triggers.”

6.4 Discriminate: remove rephrases masquerading as alternatives

Multiple outputs do not guarantee epistemic diversity. Let $C_1, \dots, C_n$ be candidate outputs and let $\sim_Q$ denote readout-equivalence for the current question: two candidates are equivalent if they license the same decisive distinctions, predictions, or actions. The useful candidate count is therefore

$$N_{\text{distinct}} = |\{C_1, \dots, C_n\} / \sim_Q|, \quad (18)$$

not n . The discrimination step asks: which candidates differ only in wording? Which introduce a new variable, causal direction, ontology, comparison, or testable consequence? This stage is inherited directly from the programme's distinction between raw candidate multiplication and readout-distinguishable alternatives [1, 2].

Prompt function: "Cluster the candidates by underlying mechanism. Remove paraphrases. Keep only alternatives that would change what evidence we seek or what action we take."

6.5 Verify: leave the AI loop, but spend verification where it matters

Candidate status changes only through resistance appropriate to the target domain. AI can locate, compare, or criticize evidence, but it should not become the sole generator and certifier of the same consequential claim.

Real-life use requires a verification budget. Define a claim c as load-bearing for the present task when its falsity would materially change at least one of

$$L(c) = \{\text{Decision, Risk, Money, Health, Legal, Publication, IrreversibleAction}\}. \quad (19)$$

When $L(c) = 1$, the claim should be exposed to at least one partially independent route:

$$V = \{\text{primary source, data, experiment, calculation, expert, independent method, world outcome}\}. \quad (20)$$

Low-stakes non-load-bearing claims may receive lighter checks. The same statement can therefore move from low to high verification burden when the intended use changes. The verification stage includes provenance typing:

$$\begin{aligned} \text{Source} &\neq \text{AI synthesis} \neq \text{Human inference} \\ &\neq \text{Unverified candidate.} \end{aligned} \quad (21)$$

Prompt function: "Which claims are load-bearing for what I am about to do? For those claims, show the primary evidence, what it actually establishes, what remains inference, and one independent route that could disagree. Do not spend the same verification effort on trivial details."

6.6 Integrate: the human reconstructs the revised model

After verification, the human—not the AI—states the revised model. A minimal integration record is

$$I_s = \langle \Delta M_s, E_s^{\text{decisive}}, U_s^{\text{remain}}, \text{Next}_s \rangle, \quad (22)$$

where ΔM_s is what changed, E_s^{decisive} the evidence or distinction that caused the change, U_s^{remain} what remains unresolved, and Next_s the next test or action.

This stage prevents a subtle failure: a user can recognize an excellent AI synthesis without being able to reconstruct the reasoning that made it excellent. Integration therefore uses generation and retrieval rather than recognition alone [21, 22].

Human action: close or hide the AI answer and explain the revised model in one's own words before asking AI to critique the reconstruction.

6.7 Remove: distinguish reset from true removal

The decisive assistance is then removed when Human Return needs to be measured. A reset breaks conversational momentum; removal tests whether the decisive capability still belongs to the human. They are not identical.

$$\text{Reset} : \text{fresh framing/session/source route}, \quad (23)$$

$$\text{Removal} : \text{absence of decisive AI assistance}. \quad (24)$$

A fresh chat can be a useful reset, but with memory, personalization, shared files, or persistent agents it may not constitute strong removal. Stronger removal includes a blank document, an offline or no-AI transfer case, a timed outage drill, or an independently completed decisive step. Coupled performance cannot reveal what belongs to the human.

6.8 Return: measure what came back with the person

The terminal readout is Human Return:

$$R_H^{\text{return}} = \langle C, T, S, A \rangle, \quad (25)$$

where C is Conceptual reconstruction, T Tool selection, S Skill execution, and A Alternative generation under unaided criteria [10]. A successful conversation need not increase all four. It must state what returned and what remained system-dependent.

A session-level conversion vector can be written

$$\Delta H_s = \langle \Delta C_s, \Delta T_s, \Delta S_s, \Delta A_s, \Delta A_{H,s}^{\text{corr}}, \Delta A_{H,s}^{\text{live}} \rangle. \quad (26)$$

The central non-collapse is

$$\Delta \text{Performance}_{\text{AI}} > 0 \not\Rightarrow \Delta H_s > 0. \quad (27)$$

Return should be sampled, not ritualized. Full removal after every interaction is unnecessary. Let

$$F^{\text{return}} = f(\text{Criticality}, \text{LearningNeed}, \text{FailureCost}, \text{DependencyRisk}). \quad (28)$$

Learning-intensive tasks may require frequent Return tests; routine automation may need only periodic fallback drills; trivial low-stakes tasks may require none. The aim is to detect loss of decisive human competence before it becomes operationally invisible.

Field evidence also cautions against assuming that AI use necessarily causes deskilling. In customer support, generative-AI assistance increased productivity and produced evidence of durable worker learning during system outages, especially among less experienced workers [29]. The protocol must therefore measure the sign of Human Return rather than presuppose it.

7 ACTION AND WORLD FEEDBACK: CLOSE THE HUMAN-DEVELOPMENT LOOP

Human Return establishes what came back with the person, but a live problem is not fully closed by verbal reconstruction. When the topic concerns action, repair, decision, or skill, the revised understanding should return to the world that generated the problem. Human Learning in the programme already treats real-problem application and feedback as part of the transition from understanding to corrigible skill [9].

Let a_s be the human-selected next action emerging from integration and let $\delta_{s+1}^{\text{world}}$ be the relevant returned record from the situation. Then

$$I_s \rightarrow a_s \rightarrow \delta_{s+1}^{\text{world}} \rightarrow H_{s+1,0}. \quad (29)$$

The returned record may confirm the revised model, expose a new residual, reveal that the problem was misframed, or create a new problem. The wider cycle is therefore

$$\begin{aligned} \text{Live Problem} &\rightarrow \text{Question} \rightarrow \text{Dialogue} \\ &\rightarrow \text{Human Return} \rightarrow \text{Action} \\ &\rightarrow \text{World Feedback} \rightarrow \text{Revision or New Problem}. \end{aligned} \quad (30)$$

This does not require every conversation to end in overt behavior. Pure learning, creative exploration, and basic inquiry can terminate at a different boundary. The claim is conditional: where the original topic was action-relevant, a conversation that never returns to the relevant world risks mistaking linguistic closure for practical resolution.

World-closure proposition [Open]. For live problems with observable consequences, dialogue policies that include post-conversation action and feedback will produce better calibration and transferable Human Return than otherwise identical policies that terminate at AI-generated synthesis.

8 EXPANSION AND CONTRACTION MUST ALTERNATE

A productive conversation should not remain in permanent expansion. The protocol therefore alternates two modes:

Expansion : maximize candidate diversity (31)

and discriminability, (32)

Contraction : prune by evidence, provenance, (33)

stakes, and action relevance. (34)

Permanent expansion creates overload and rhetorical abundance; premature contraction creates tunnel risk. The switch should occur when additional candidates no longer create a new equivalence class or when a load-bearing decision now requires world-side evidence.

9 STOP RULES AND RESET RULES

A conversation should stop or reset when conversational momentum is increasing faster than evidence. The following are candidate stop conditions:

- three or more turns produce no new non-equivalent candidate class;
- confidence rises while provenance or external evidence does not;
- AI begins validating a claim primarily by reference to its own earlier synthesis;
- the user cannot restate the current reasoning without reopening the transcript;
- the task becomes high-stakes enough that an independent expert/source is required;
- the user is seeking confirmation rather than discrimination;
- objections are repeatedly absorbed into one favored frame without changing its predicted consequences.

A reset can take four forms: fresh-session restart, prompt inversion, independent-source check, or human-only reconstruction. The purpose is not variation for its own sake; it is to break local conversational path dependence and test whether the current conclusion survives altered framing.

10 DIALOGUE POLICY MUST DEPEND ON THE GOAL

One universal prompt policy is unlikely to maximize human conversion. Let

$$\pi^{\text{dialogue}} = f(\text{TopicMode, Goal, Skill, Stakes, Domain, LearningNeed, Reversibility}). \quad (35)$$

Table 2 specifies candidate policies and recommended deployment modes.

10.1 Personal and relational dialogue requires a different verification logic

In interpersonal life, a “primary source” may be unavailable because the unknown is another person’s intention, feeling, or motive. The danger is interpretive reinforcement: AI can make one user’s story more coherent without gaining privileged access to the absent person’s mind. The recommended route is

$$\begin{aligned} \text{Experience} &\rightarrow \text{Interpretations} \rightarrow \text{Absent Perspective} \\ &\rightarrow \text{Observable Evidence} \\ &\rightarrow \text{Direct Human Conversation.} \end{aligned} \quad (36)$$

A useful prompt is: “Separate what I directly observed from my interpretation. Give alternative explanations that do not accuse the absent person without evidence. What question should I ask them directly before concluding?” This mode treats AI as an interpretation-expander, not an oracle of hidden motives.

11 PROTOCOL BURDEN AND BEHAVIORAL ADOPTION

An epistemically protective protocol can fail socially if its burden causes users to ignore it. Define a usability-burden vector

$$B_{\text{use}} = \langle \text{Time}, \text{CognitiveLoad}, \text{VerificationCost}, \text{Interruption}, \text{LiteracyDemand} \rangle. \quad (37)$$

Cognitive-forcing work already shows a friction-performance preference trade-off: interventions that reduce overreliance can demand more effort and may be less preferred [16, 19]. Therefore

$$\text{Epistemically optimal} \not\equiv \text{Behaviorally adoptable.} \quad (38)$$

A deployable protocol should report not only Human Return but completion rate, abandonment, time cost, satisfaction, repeated-use willingness, and whether benefits persist under ordinary time pressure. DCP-Lite exists specifically to lower B_{use} where a full protocol would add little value.

12 PROVENANCE LEDGER: WHAT IS THIS SENTENCE DOING?

A compact dialogue ledger can prevent candidate forgetting. Each important statement receives one status:

$$\text{Status} \in \{\text{Source}, \text{AI-Synthesis}, \text{Human-Inference}, \text{Candidate}, \text{Decision}\}. \quad (39)$$

A candidate may be useful without being true; an AI synthesis may accurately summarize sources without adding independent evidence; a decision may be justified under uncertainty without converting uncertainty into fact. This ledger operationalizes the programme’s claim-level non-collapse discipline and makes later auditing possible [1].

13 WHAT SHOULD THE HUMAN SAY TO AI? ADAPTIVE OPERATIONAL SCRIPTS

13.1 Topic Entry: before any protocol

For capability-sensitive use, begin with one question: “What unresolved difference in my present life, work, understanding, or environment deserves my attention now?” If the purpose is curiosity or play, state that explicitly: “I am exploring, not solving a current problem; help me widen the field without pretending there is an urgent problem.” This makes the agenda visible before AI begins optimizing within it.

13.2 The one-minute DCP-Lite script

For ordinary use, four speech acts are sufficient:

1. State: "This is the problem or goal I chose to work on; this part matters most."
2. Challenge: "What obvious alternative, exception, or mistake might I be missing?"
3. Check: "Which fact would materially change my action if it were wrong, and how should I verify it?"
4. Own: "Summarize the decision boundary, not just the answer. I will make the final choice."

13.3 The full DCP-Standard script

When learning, research, transfer, or future unaided competence matters:

1. Topic/Anchor: “This is the live problem or explicit exploration goal I chose. This is my current model, uncertainty, and what would change my mind. If I lack a model, help me clarify without silently replacing my agenda.”
2. Expand: “Give me non-equivalent routes, not rephrasings. State what changes structurally in each.”
3. Oppose/Boundary-Test: “If the issue is genuinely open, build the strongest rival. If evidence is strongly one-sided, identify boundaries, uncertainty, and revision triggers instead of manufacturing an opposite.”
4. Discriminate: “Which alternatives are genuinely different in mechanism, prediction, or action? Collapse duplicates.”
5. Verify: “Which claims are load-bearing for my intended use? Verify those with primary evidence, provenance, uncertainty, and an independent route.”
6. Integrate: “Do not summarize yet. I will restate what changed; then critique my reconstruction.”
7. Remove: “I am removing decisive AI assistance and will attempt the critical step independently.”
8. Return: “Test what I can reconstruct, select, execute, and generate without your decisive help.”

13.4 DCP-Critical escalation rule

When the consequence becomes high-stakes or hard to reverse, the protocol adds: “Do not act as the sole certifier. Identify the independent professional, primary record, calculation, test, or institutional route required before action.” The critical mode

Table 2: Task-contingent dialogue policies and deployment levels.

Goal	Mode	AI should mainly do	Human must retain	Protocol emphasis
Routine / admin	Lite	automation, formatting, ordinary translation, repetitive operations	exception detection, ownership of consequential wording, fallback for critical functions	State–Challenge–Check–Own; no routine
Learning	Standard	hints, examples, error diagnosis, graduated counterexamples, retrieval prompts	first attempt, explanation, retrieval, transfer, decisive skill step	Remove/Return delay full solution when learning value is high; frequent Return
Research	Standard/Critical	search, literature transport, rival generation, synthesis, code assistance	problem ownership, source judgment, validation design, interpretation, claim strength	provenance, non-equivalent rivals, primary-source verification
Decision support	Standard/Critical	structured alternatives, sensitivity analysis, counterfactuals	initial judgment, value trade-offs, accountability, override competence	pre-AI estimate when feasible; verification budget; escalate with stakes
Creative work	Lite/Standard	divergence, recombination, style variation, analogy	purpose, taste, selection, responsibility, authorship claim	expansion first; lighter truth burden until factual claims become load-bearing
Personal / relational	Lite/Standard	separate facts from interpretations, generate charitable alternatives, surface absent perspectives, draft questions	ownership of feelings/values, direct communication, knowledge that the other person’s mind is not directly observed	Facts $\neq$ interpretation; avoid diagnosing absent people; return to observable evidence and human conversation
High-stakes domains	Critical	evidence organization, differential/rival generation, checklists	accountable professional judgment and independent verification	no single-route certification; expert/world validation before action

terminates in accountable human/world-side validation, not in a more confident AI answer.

14 FORMAL SUCCESS CONDITION

Let P_s^{assist} be current assisted performance, H_s the Human Return/agency vector after removal, and B_{use}^s the usability-burden vector. A dialogue policy π is a high-conversion policy for a declared task if

$$\pi^* \in \arg \max_{\pi} \mathbb{E}[\Delta H_s \mid \pi, \text{task}, \text{user}], \quad (40)$$

subject to minimum task-performance and safety constraints, a declared burden/adoption ceiling, and—for human-development use—a visible topic-ownership condition

$$B_{\text{use}}(\pi) \leq \bar{B}(\text{task}, \text{user}). \quad (41)$$

This differs from

$$\pi^{\text{perf}} \in \arg \max_{\pi} \mathbb{E}[P_s^{\text{assist}} \mid \pi], \quad (42)$$

and the two policies need not coincide.

Dialogue Conversion Proposition [Open]. For tasks in which learning, judgment transfer, or future unaided competence matters, the assistance policy that maximizes immediate performance will often differ from the policy that maximizes delayed Human Return. **Deployment Proposition [Open].** The full protocol will not maximize real-world adoption across all tasks; adaptive triage should achieve similar protection at

lower burden for low- and medium-stakes use. **Agenda Proposition [Open].** When the objective is human development, self-selected live-problem entry will often produce greater relevance, action transfer, and Human Return than an otherwise comparable AI-suggested agenda, after baseline motivation and task difficulty are controlled.

15 EMPIRICAL PROGRAMME

A tier-one test should compare protocols, not merely AI-versus-no-AI.

15.1 Multi-arm randomized design

Randomize or adaptively assign participants to: (A) direct-answer AI; (B) DCP-Lite; (C) explanation/cognitive-forcing AI; (D) adversarial-rival AI; (E) DCP-Standard; (F) DCP-Critical when stakes justify escalation; and (G) no-AI baseline. Tasks should include at least one learning task, one decision task, one open-ended research/creation task, one routine task, and one personal/relational interpretation task with controlled ground truth or follow-up evidence where ethically feasible. A crossed topic-entry factor should compare (i) self-selected live problem, (ii) AI-suggested topic matched for domain and difficulty, and (iii) curiosity-led self-selected exploration.

Primary outcomes are measured at three times: immediate assisted performance, delayed unaided Human Return, and transfer to an unseen/held-out case. Deployment outcomes are co-primary for real-life validity: completion rate, time cost, cognitive load, satisfaction, abandonment, repeated-use

willingness, over/under-reliance, and calibration. The study should test whether Lite preserves most of the benefit of Standard in low-stakes tasks and whether Critical adds value only where stakes or irreversibility justify its burden.

15.2 Primary hypotheses

H1. Direct-answer AI will often maximize immediate performance but not delayed Human Return in learning-intensive tasks. **H2.** DCP and cognitive-forcing conditions will reduce overreliance but may increase cognitive load. **H3.** DCP will produce more non-equivalent unaided alternatives at follow-up than direct-answer AI. **H4.** Provenance accuracy and error detection after AI removal will mediate the relationship between protocol and Human Return. **H5.** Effects will be heterogeneous by prior skill: novices may require more scaffolded opposition/verification than experts. **H6.** In creation tasks, aggressive expansion may improve coupled output without harming Return, whereas high-stakes decision tasks will benefit more from early anchoring and independent verification. **H7.** DCP-Lite will achieve higher adoption and lower burden than DCP-Standard in low-stakes tasks with little loss of safety or ownership. **H8.** In relational interpretation tasks, facts-versus-interpretation separation and absent-perspective prompts will reduce unwarranted certainty about another person's motives compared with direct-answer AI. **H9.** For capability-sensitive tasks, self-selected live-problem entry will increase perceived relevance, follow-through, and delayed Human Return relative to matched AI-suggested topics, while curiosity-led exploration will outperform problem-first on breadth and intrinsic exploration outcomes. **H10.** For action-relevant live problems, Action → World Feedback closure will improve calibration and transfer relative to otherwise identical dialogue that terminates at synthesis.

15.3 Falsification and removal conditions

The protocol should be simplified if: (i) ordering the stages adds no value beyond generic “think critically” instructions; (ii) Human Return does not predict later independent performance beyond standard skill tests; (iii) non-equivalent alternative counting adds no value over raw idea count; (iv) reset or independent-route interventions do not reduce tunnel effects; (v) direct-answer assistance consistently improves both immediate and delayed outcomes with no detectable agency or dependence cost; (vi) Lite/Standard/Critical triage adds no value over one fixed protocol; or (vii) protocol burden is so high that protective effects disappear under realistic adoption and time constraints; (viii) live-problem topic entry adds no value beyond motivation, relevance, and task matching; or (ix) action/world-feedback closure adds no predictive or calibration value beyond Human Return alone.

16 URGENCY AND THE REVERSIBILITY WINDOW

The protocol is not merely pedagogical. Repeated AI use can become infrastructural habit before institutions know what human capacities have been lost, and AI-supplied agendas can become normalized before users notice that topic selection itself has shifted outside their control. The Human Conversion Imperative therefore adds a Reversibility Principle: when machine capability changes faster than human adaptation and the recovery cost of lost skills or routes rises, preserving independent competence earlier may be cheaper than rebuilding it later [12].

At the dialogue level, this means sampling whether the human can still perform the decisive cognitive operation without the system at a frequency proportional to learning need, failure cost, and dependency risk. At the organizational level, it means maintaining fallback expertise, exportable records, independent validation routes, and scheduled outage or transfer drills for critical functions. The aim is not to freeze pre-AI work. It is to prevent irreversible dependence from being mistaken for progress while avoiding needless friction in trivial work.

17 LIMITS

The adaptive protocol is intentionally strong but not universal. First, the triage rules are conceptual and not yet calibrated: the boundary between Lite, Standard, and Critical requires empirical thresholds. Second, problem-first can itself fail: people may misidentify the problem, ruminate on a salient but low-value concern, or inherit a problem frame from family, institutions, markets, or the AI itself. The Topic Entry layer therefore protects agenda visibility, not infallible self-knowledge. Third, problem-first must not suppress curiosity, art, play, basic science, or exploratory learning; the Open Exploration mode is constitutive, not an exception added for convenience [28]. Fourth, friction can produce under-reliance as well as healthy skepticism, and users differ in tolerance for cognitive effort [16, 19]. Fifth, Human Return is task-relative; losing a low-value manual procedure may be rational if higher-level capability improves. Sixth, “removal” is difficult to standardize in systems with memory, personalization, agents, or organizational integration. Seventh, independent verification can itself be biased, inaccessible, expensive, or slower than the decision window. Eighth, relational dialogue can reduce premature certainty but cannot infer an absent person's private mental state. Ninth, the protocol does not solve misinformation, professional judgment, or institutional power by procedure alone. Tenth, minimum sufficient friction and problem-first entry are design hypotheses: if lighter protocols or alternative topic-selection policies consistently preserve equal or greater Human Return, the architecture should be simplified.

18 CONCLUSION

The question “How should humans converse with AI?” begins one step earlier than prompting. In an AI-rich environment, a system can shape not only answers but the agenda of attention. The first design question is therefore: what deserves to become the topic of dialogue, and who owns that choice?

For human-development use, this paper proposes a default of Problem-First Dialogue: begin from a live, presently consequential unresolved difference in the person’s life, work, understanding, or environment. The principle is not universal. Curiosity, play, art, basic inquiry, and routine delegation remain legitimate entry modes. The non-collapse is explicit:

Problem-First $\neq$ Problem-Only. (43)

The complete adaptive architecture is

Topic Entry $\rightarrow$ Triage $\rightarrow$ Minimum Sufficient Dialogue
 $\rightarrow$ Human Return $\rightarrow$ Action/World Feedback
 $\rightarrow$ Revision or New Problem. (44)

For low-stakes use, DCP-Lite asks the human to State, Challenge, Check, and Own. When learning, research, judgment transfer, or future competence matters, the canonical protocol remains

Anchor $\rightarrow$ Expand $\rightarrow$ Oppose/Boundary-Test
 $\rightarrow$ Discriminate $\rightarrow$ Verify $\rightarrow$ Integrate (45)
 $\rightarrow$ Remove $\rightarrow$ Return.

When stakes become high, independent world-side or professional validation becomes mandatory.

Humans should use AI aggressively where machines are strong—breadth, search, translation, recombination, simulation, and candidate generation—while retaining the functions that make human development possible: agenda/problem ownership, world contact, normative responsibility, integration, refusal, correction, and fallback competence. A high-quality conversation is not one in which AI appears maximally intelligent, nor one in which the protocol appears maximally rigorous. It is one in which the user spends attention on a topic they can still recognize as theirs, uses only the friction the task deserves, leaves with more durable capability, and—where the problem is action-relevant—lets the world answer back.

The decisive practical questions are therefore:

What unresolved difference deserves my attention?
 How much AI/friction does this task require?
 What remains with me after AI is removed? (46)
 What does the world return when I act?

REFERENCES

- [1] Yaoharee Lahtee. Epistemic fusion or epistemic tunnel: A phenomenology-anchored, global-literature-constrained, problem-first architecture of human-ai collaboration (epistemic note v8.1 – strong-claim / legitimate-defeater edition). Human-AI Readout Programme, v8.1, 5 September 2026, 2026. <https://doi.org/10.5281/zenodo.22331922>.
- [2] Yaoharee Lahtee. Choice begins before choice: Meaning-shaped accessibility, live possibility, and effective agency in human-ai systems. Human-AI Readout Programme, Full Paper v1.0 Strong Claim, 5 September 2026, 2026. <https://doi.org/10.5281/zenodo.22357788>.
- [3] Yaoharee Lahtee. When ai expands human potential: Reflective dissonance, epistemic agency, and constraint. Open Civil Science Initiative preprint, 25 March 2026, 2026. Human-AI Readout Programme. <https://doi.org/10.5281/zenodo.19215748>.
- [4] Yaoharee Lahtee. The readout condition: Distinguishability, access, and epistemic warrant. Human-AI Readout Programme, elevated full paper, August 2026, 2026. <https://doi.org/10.5281/zenodo.22301318>.
- [5] Yaoharee Lahtee. From problem to hypothesis: Dynamic semantic mobility, bounded knowers, and the readout-discriminable hypothesis space. Human-AI Readout Programme, GLOSA K0 final author draft, 4 September 2026, 2026. <https://doi.org/10.5281/zenodo.22307148>.
- [6] Yaoharee Lahtee. The epistemic chain reaction. Human-AI Readout Programme, K0-2026-09-04-v3, 2026. <https://doi.org/10.5281/zenodo.22308072>.
- [7] Yaoharee Lahtee. Experience is the human lora. Human-AI Readout Programme, 2026. <https://doi.org/10.5281/zenodo.21425420>.
- [8] Yaoharee Lahtee. Ai, translation, and access to event-specific context. Human-AI Readout Programme, 2026. <https://doi.org/10.5281/zenodo.18517054>.
- [9] Yaoharee Lahtee. Human learning as epistemic architecture. Human-AI Readout Programme, revised v4 preprint, 2026. <https://doi.org/10.5281/zenodo.22341297>.
- [10] Yaoharee Lahtee. Ctsa human-return readout: A session-boundary measurement architecture for retained human capability after ai. Human-AI Readout Programme, v9.1 FINAL, 5 September 2026, 2026. <https://doi.org/10.5281/zenodo.22339909>.
- [11] Yaoharee Lahtee. After labour: Human position in an ai-robotic world system. Human-AI Readout Programme, full world-system standalone draft v1.1, 6 September 2026, 2026. Companion paper, same deposit batch as the present paper; DOI 10.5281/zenodo.22481924.

- [12] Yaoharee Lahtee. The human conversion imperative: Machine acceleration, human potential, and the reversibility window in an ai-robotic world. Human-AI Readout Programme, standalone proposal v1.0, 6 September 2026, 2026. Companion paper, same deposit batch as the present paper; DOI 10.5281/zenodo.22481926.
- [13] Roni Reiter-Palmon. The role of problem construction in creative production. *Journal of Creative Behavior*, 51(4):323–326, 2017. doi: 10.1002/jocb.202.
- [14] Stefano Magistretti, Christophe T. A. Pham, and Claudio Dell’Era. The creative process of problem framing for innovation: An integrative review and research agenda. *Journal of Product Innovation Management*, 42(6):987–1018, 2025. doi: 10.1111/jpim.12783.
- [15] Michelle Vaccaro, Abdullah Almaatouq, and Thomas W. Malone. When combinations of humans and AI are useful: A systematic review and meta-analysis. *Nature Human Behaviour*, 8:2293–2303, 2024. doi: 10.1038/s41562-024-02024-1.
- [16] Zana Bućinca, Maja Barbara Malaya, and Krzysztof Z. Gajos. To trust or to think: Cognitive forcing functions can reduce overreliance on AI in AI-assisted decision-making. In *Proceedings of the ACM on Human-Computer Interaction (CHI)*, 2021. arXiv:2102.09692.
- [17] Helena Vasconcelos, Matthew Jorke, Madeleine Grunde-McLaughlin, Tobias Gerstenberg, Michael S. Bernstein, and Ranjay Krishna. Explanations can reduce overreliance on AI systems during decision-making. *Proceedings of the ACM on Human-Computer Interaction*, 7(CSCW1):129:1–129:38, 2023. doi: 10.1145/3579605.
- [18] Sander de Jong, Vera Paananen, Benjamin Tag, and Niels van Berkel. Cognitive forcing for better decision-making: Reducing overreliance on AI systems through partial explanations. *Proceedings of the ACM on Human-Computer Interaction*, 9(2):CSCW048, 2025. doi: 10.1145/3710946.
- [19] Hui Yi Tian, Hamza Amin, and Ming Yin. Understanding the effects of AI-assisted critical thinking on human-AI decision making. In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems*, 2026. doi: 10.1145/3772318.3790785. Article 810.
- [20] Norman J. Slamecka and Peter Graf. The generation effect: Delineation of a phenomenon. *Journal of Experimental Psychology: Human Learning and Memory*, 4(6):592–604, 1978. doi: 10.1037/0278-7393.4.6.592.
- [21] Henry L. Roediger, III and Andrew C. Butler. The critical role of retrieval practice in long-term retention. *Trends in Cognitive Sciences*, 15(1):20–27, 2011. doi: 10.1016/j.tics.2010.09.003.
- [22] Jeffrey D. Karpicke and Janell R. Blunt. Retrieval practice produces more learning than elaborative studying with concept mapping. *Science*, 331(6018):772–775, 2011. doi: 10.1126/science.1199327.
- [23] Manu Kapur. Productive failure. *Cognition and Instruction*, 26(3):379–424, 2008. doi: 10.1080/07370000802212669.
- [24] Manu Kapur. Productive failure in learning math. *Cognitive Science*, 38(5):1008–1022, 2014. doi: 10.1111/co gs.12107.
- [25] Hamsa Bastani, Osbert Bastani, Alp Sungu, et al. Generative AI without guardrails can harm learning: Evidence from high school mathematics. *Proceedings of the National Academy of Sciences*, 122(26):e2422633122, 2025. doi: 10.1073/pnas.2422633122.
- [26] Moshe Glickman and Tali Sharot. How human-AI feedback loops alter human perceptual, emotional and social judgements. *Nature Human Behaviour*, 9:345–359, 2025. doi: 10.1038/s41562-024-02077-2.
- [27] Zana Bucinca, Siddharth Swaroop, Amy E. Paluch, Susan A. Murphy, and Krzysztof Z. Gajos. Towards optimizing human-centric objectives in AI-assisted decision-making with offline reinforcement learning, 2024. arXiv:2403.05911.
- [28] Francesco Poli, Jill X. O’Reilly, Rogier B. Mars, and Sabine Hunnius. Curiosity and the dynamics of optimal exploration. *Trends in Cognitive Sciences*, 28(5):441–453, 2024. doi: 10.1016/j.tics.2024.02.001.
- [29] Erik Brynjolfsson, Danielle Li, and Lindsey Raymond. Generative AI at work. *Quarterly Journal of Economics*, 140(2):889–942, 2025. doi: 10.1093/qje/qjae044.